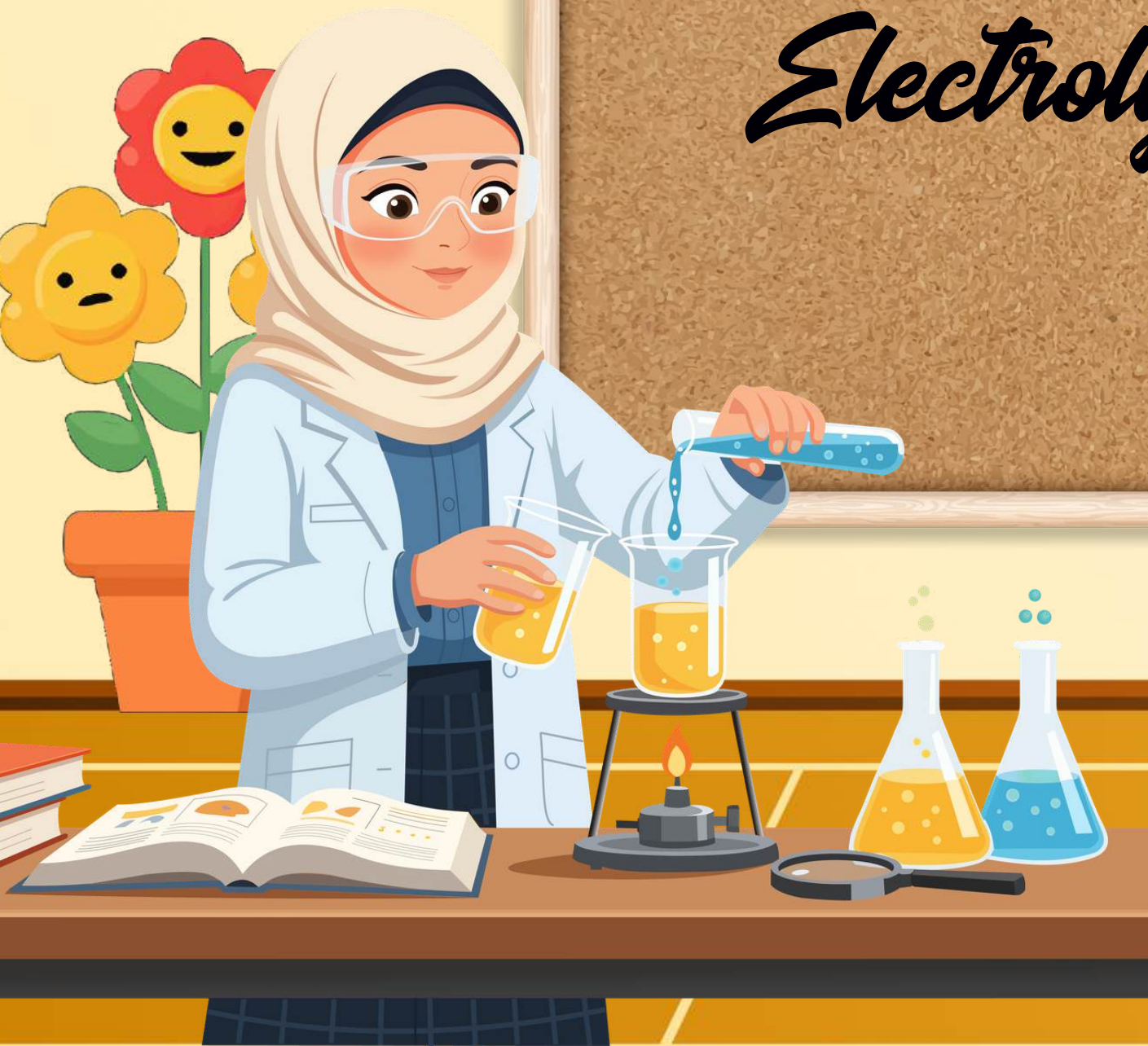




Electrolysis: Splitting Water with Electricity





What if Electricity Could BREAK Water?

Imagine this: What if electricity could actually BREAK water apart? Look at the bubbles forming in the glass! What do you think is happening inside? Let your imagination run wild and think about the magic of science!





Let's Think Together!

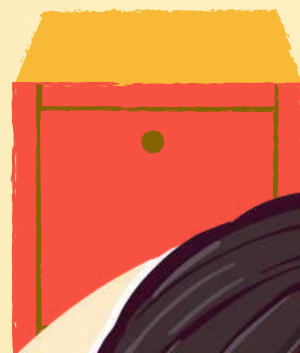
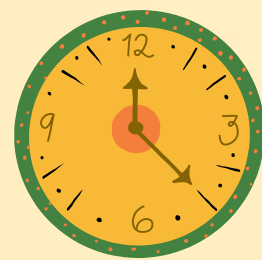
Kitchen Electricity?

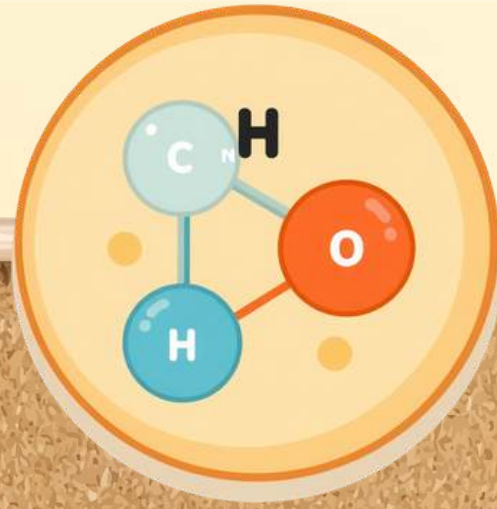


What's in Water?



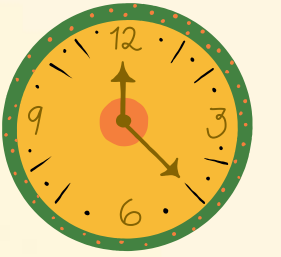
Why Bubbles?





Water's Secret Recipe

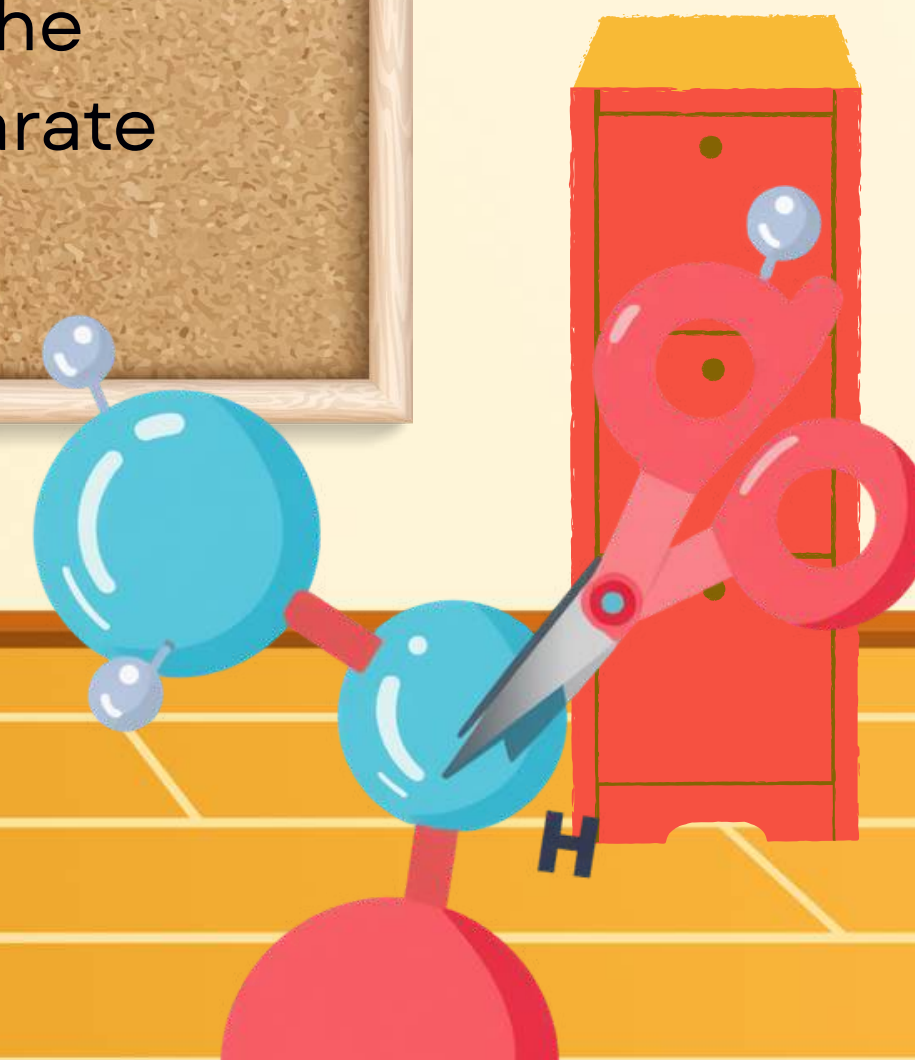
Water is made of tiny parts called molecules.
Each water molecule has 2 Hydrogen + 1
Oxygen. Think of it like roti dough made of atta
+ water! The special name for water is H_2O .

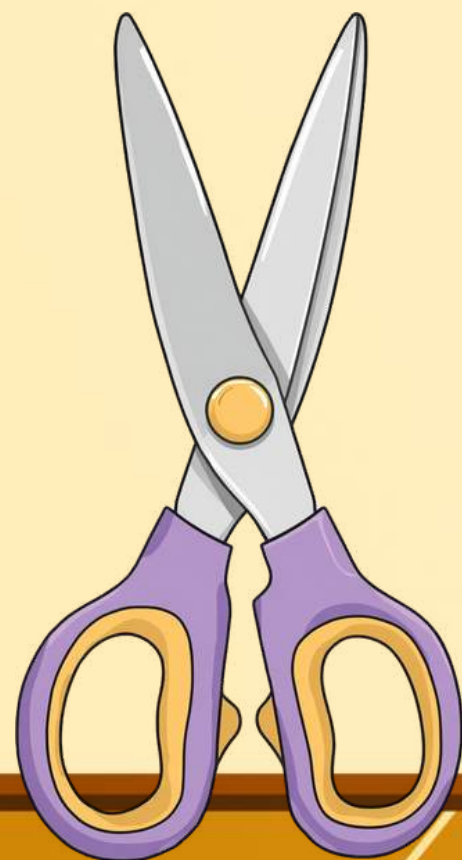




Breaking the Water Molecule

Electrolysis means using electricity to 'cut' water into hydrogen and oxygen gases. Imagine scissors cutting the bonds between atoms – like cutting rakhis! The electricity breaks apart the H_2O molecule into separate hydrogen (H_2) and oxygen (O_2) gases.





What is Electrolysis?

Electrolysis is Like Scissors Cutting a Knot!
Electricity Breaks Water into Gases



ELECRIDE

Electrodes – Our Pencils Are Important!

Electrodes are two pencils that let electricity in and out of the water. Think of them like dosa tawa handles – one is positive (+) and one is negative (-). The electricity flows through these pencils to split the water!

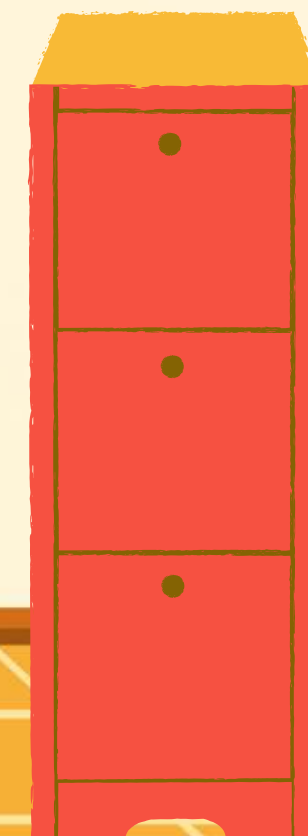
Key Term: Electrode – a conductor that electricity enters or leaves through.





Electrolyte – Salt Helps!

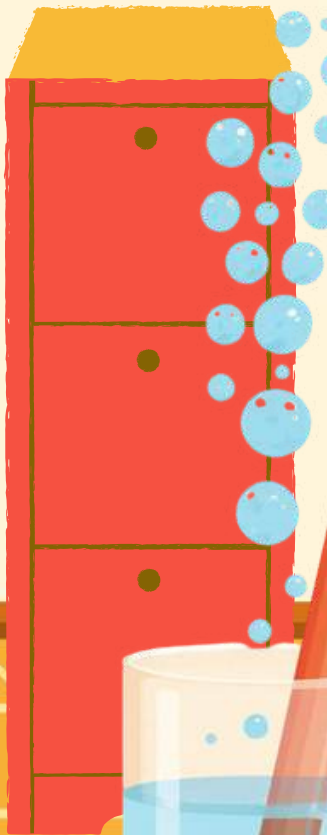
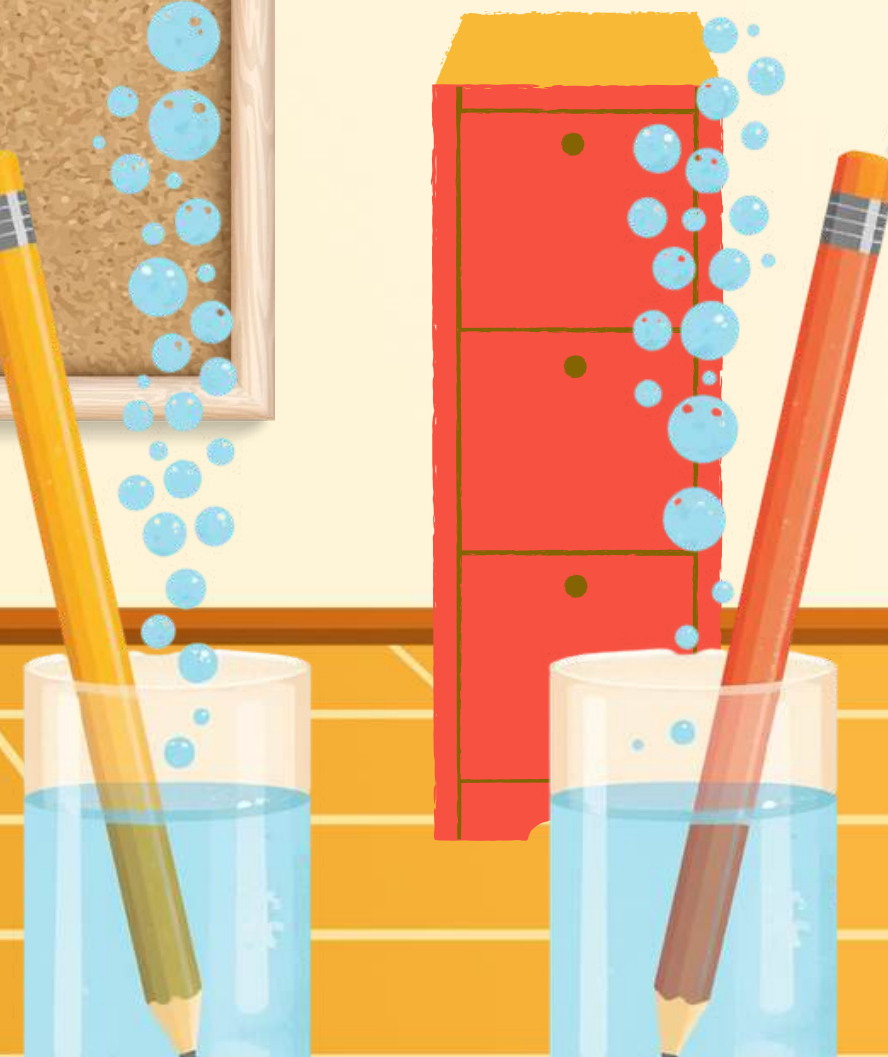
Salt makes water conduct electricity better. Think of it like adding Add salt to lemon water to make it taste just right! The salt helps electricity flow through the water easily. This special helper is called an "Electrolyte."





What Happens at Electrodes?

At the negative electrode, hydrogen gas forms with MORE bubbles! At the positive electrode, oxygen gas forms with fewer bubbles. Watch carefully – the pencil with more bubbles is making hydrogen (H_2), and the one with less bubbles is making oxygen (O_2)!





Let's Do the Experiment!

Glass of Water

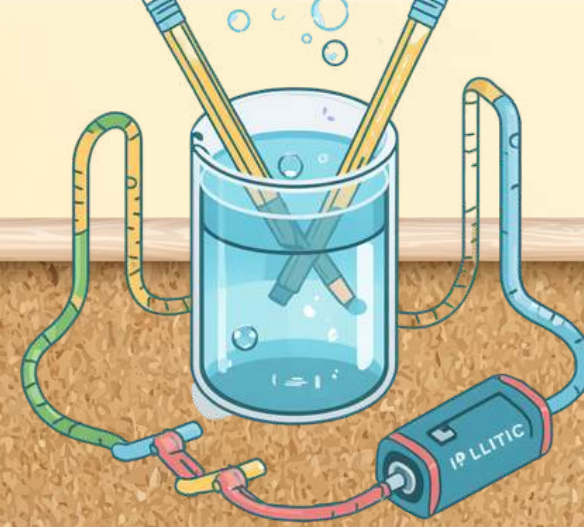
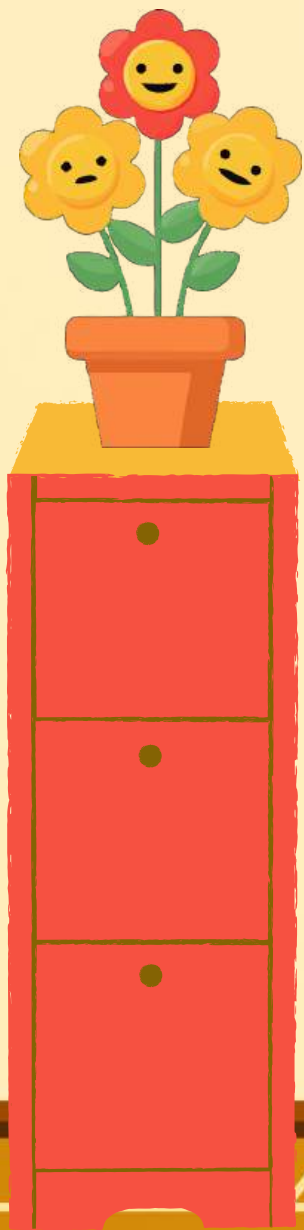


9V Battery



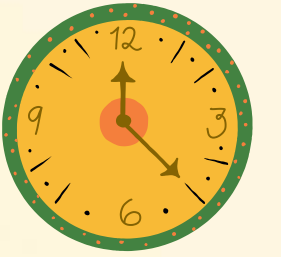
2 Pencils





Step-by-Step Experiment

- 01 Fill glass with water
- 02 Add salt and stir well
- 03 Connect wires to battery
- 04 Dip pencils in water





Safety First!

Use 9V Battery Only

Adult Supervision

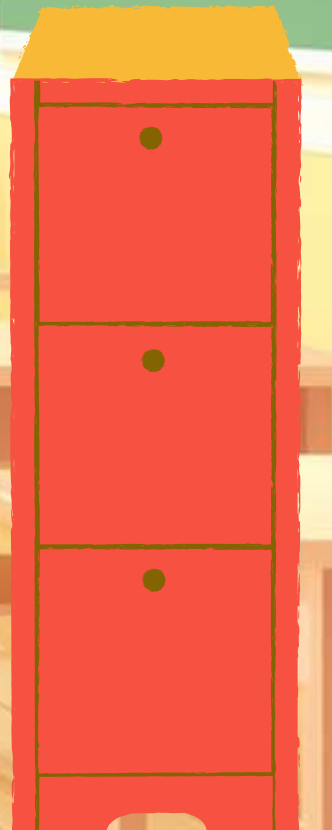
Don't Touch Wires

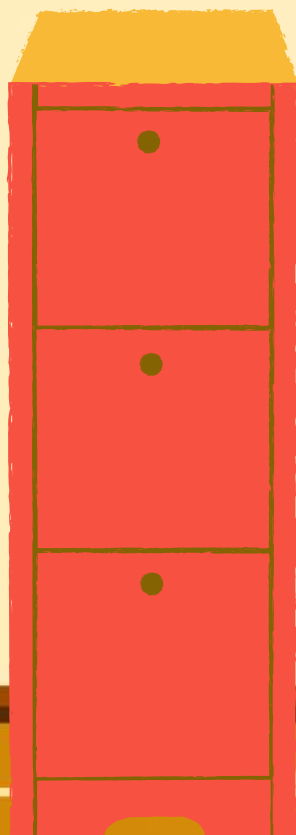




Real-World Uses of Electrolysis in India

Electrolysis helps us every day! Green hydrogen buses run in Delhi and Mumbai. Hospitals use oxygen cylinders made by electrolysis. Gold jewellery for Indian weddings gets its shine from electroplating. Even INS Arihant submarine makes oxygen underwater!



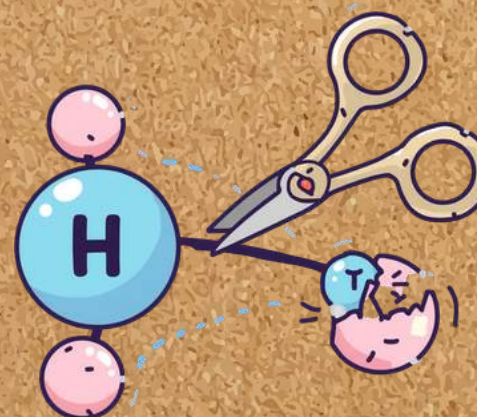


Let's Recap!

Water = H_2O



Electrolysis



Electrodes





Quick Quiz Time!

Q1

What does electrolysis split?

a) Milk b) Water c) Salt

Q2

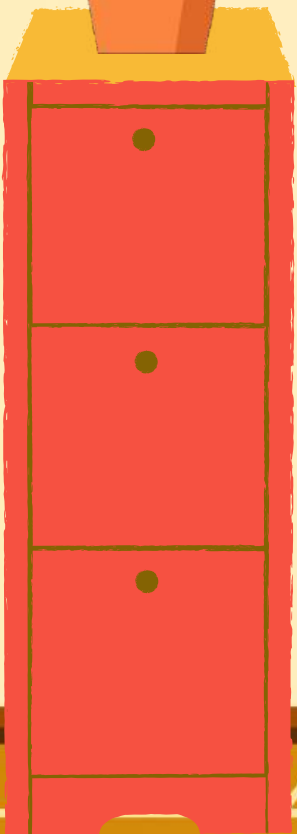
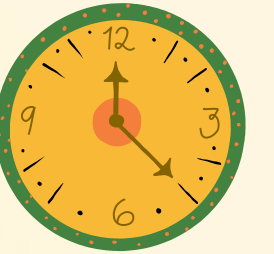
What gas forms at the negative electrode?

a) Oxygen b) Hydrogen c) Carbon dioxide

Q3

Why do we add salt to the water?

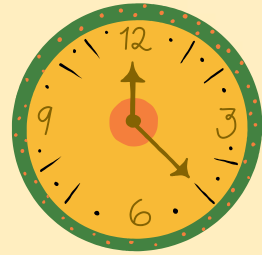
a) For taste b) To conduct electricity
c) To colour water





Your Homework Challenge!

Ask a parent or elder: Have you seen gold plating done?
Find out where in your city jewellers use electrolysis! Then
draw what you imagine the setup looks like. Be creative and
share your discoveries with the class!





Thank You!
Keep exploring the magic of science!

